Journal of the Royal Statistical Society, Series C (Applied Statistics), , 1–12
doi:
Advance Access Publication Date: Day Month Year
Paper

Supplementary material for "Spatio-temporal generation of precipitation using a Hidden Markov Model, extended extreme distributions and conditional Gaussian fields"

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Associate Editor: Name

FOR PUBLISHER ONLY Received on Date Month Year; revised on Date Month Year; accepted on Date Month Year

Abstract

1. Additional information on the statistical model

1.1. Illustration of the correlated binary-valued distribution

The emission distribution in the HMM model presented in the main article relies on a correlated binary occurrence vector $\boldsymbol{Y} = (Y_1, ..., Y_D)$, which is generated in two steps by

$$X_1, ..., X_D \sim \mathcal{N}(0, C_Y), \qquad \forall s,\ Y_s = \begin{cases} 1 & \text{if } X_s \leq \Phi^{-1}(\lambda_s) \\ 0 & \text{else} \end{cases}, \tag{1}$$

with Φ the cumulative distribution function (CDF) of the standard normal distribution.

Figure 1 illustrates examples of such binary fields for constant threshold λ and different values of the spatial scale parameter ρ in the exponential covariance $C_Y(h) = \exp(-h/\rho)$. Black pixels represent $Y_s = 1$ (rain) and white pixels represent $Y_s = 0$ (no rain). For very small ρ (e.g., $\rho = 0.01$), the field resembles independent Bernoulli draws. As ρ increases, larger and smoother spatial patterns emerge.

1.2. Overview of the model parameters

Figure 2 summarises the different parameters involved in thegenerator and presented in the model definition section, together with the processes to which they are related. The key and novel point compared to other strategies that model occurrences and intensity independently, is that both processes are related according to the rain probabilities $\lambda^{(n)}_{k,s,y_s^{(n-1)}}$.

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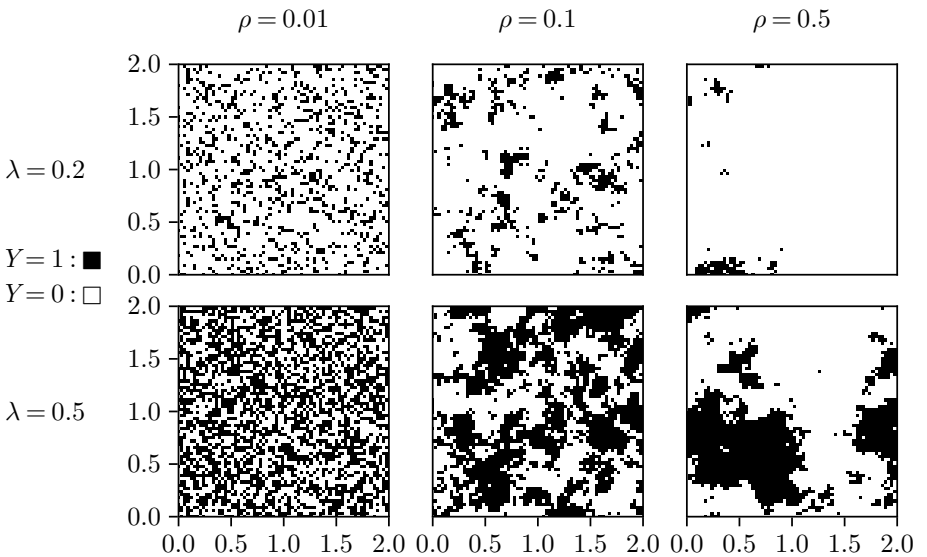

Figure 1: Generations of a binary correlated field for constant values of $\lambda = \mathbb{P}(Y_s = 1)$ over space and different values of spatial scale ρ in an exponential covariance function $C_Y(h) = \exp(-h/\rho)$.

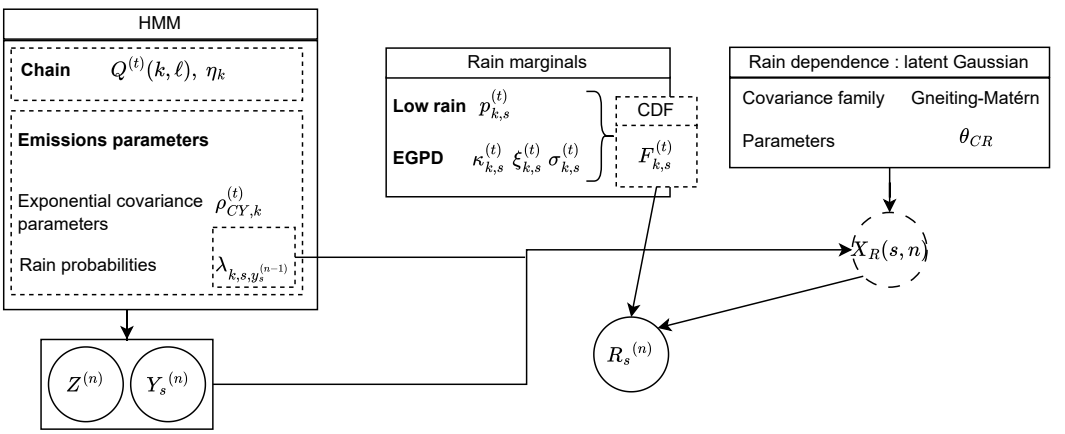

Figure 2: Outline of the model parameters, grouped into occurrence (HMM), intensity marginals and dependence parameters. The parameters are linked to the variables they influence.

2. Details on the computational choices

2.1. Integral computation methods

Computations of the likelihood L_Y within the E-step of the EM algorithm require efficient evaluation of multivariate normal integrals, whereas the maximisation of the pairwise likelihood $L_{Y,wp}$ in the M-step only requires bivariate integrals. These bivariate integrals must be computed efficiently enough to allow optimisation and, ideally, automatic differentiation.

For the multivariate case, the algorithm of Genz (1992) is employed. This method uses efficient Quasi-Monte Carlo sampling and variable transformations to achieve high accuracy and provides a parameter to control the approximation error through the number of Monte Carlo samples. More recent approaches

approximate the target distribution using bivariate or trivariate conditioning (see, for instance, Bhat (2018) and references therein), but they lack an explicit error-control parameter and tend to produce larger errors for highly correlated data, making them unsuitable for the present application. Another alternative could have been to adopt block covariance decomposition methods, such as those proposed by Genton et al. (2018), which are designed to efficiently compute the multivariate CDF for very high-dimensional problems ($D \sim 10^4$). However, this approach was not implemented because the current setting involves a moderate spatial dimension with numerous temporal points rather than extremely high-dimensional spatial data.

In the bivariate case, Lin (1995); Genz (2004); Tsay and Ke (2023) have developed efficient analytic approximations of the integral, with the most recent approach offering the best performance: it is accurate both in the central region and in the tails and often matches or outperforms existing methods. The approximation error remains small even for high correlations, and the resulting expression is easily differentiable.

2.2. Integral simplifications

2.2.1. Reformulation of integral problems into CDF of multivariate normal

The integrals involved in the likelihoods of the HMM have one finite and one infinite bound in each dimension, depending on the observed value of precipitation occurrence. They can be rewritten as the CDF of a multivariate normal distribution to take advantage of efficient methods.

Bivariate case

In the bivariate case, any integral of the pdf f_ρ of normal $\mathcal{N}\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}1 & \rho\\ \rho & 1\end{bmatrix}\right)$ can be written as a function of the bivariate normal cumulative distribution function Φ_2 and the univariate normal cumulative distribution function Φ

$$\int_{-\infty}^{a}\int_{-\infty}^{b} f_\rho = \Phi_2(a,b,\rho) \tag{2}$$

$$\int_{a}^{\infty}\int_{-\infty}^{b} f_\rho = \Phi(a) - \Phi_2(a,b,\rho) \tag{3}$$

Multivariate case

Let $R(B) = \prod_{i=1}^{D} I_i$ with $I_i = (-\infty, b_i)$ or (b_i, ∞) the integration region in the multivariate case, and f_Σ the pdf of multivariate $\mathcal{N}(0,\Sigma)$.

Let

$$D = \mathrm{diag}(d_1,\dots,d_D), \qquad d_i = \begin{cases} -1, & \text{if the bounds are } (b_i,\infty),\\ 1, & \text{if the bounds are } (-\infty,b_i),\end{cases}$$

and define

$$\tilde{b}_i = d_i b_i, \qquad \tilde{\Sigma} = D\Sigma D.$$

Then the mixed-bound integral can be rewritten as

$$\int_{R(B)} f_\Sigma(u)\,du = \int_{-\infty}^{\tilde{b}_1}\cdots\int_{-\infty}^{\tilde{b}_D} f_{\tilde{\Sigma}}(v)\,dv = \Phi_{\tilde{\Sigma}}(\tilde{b}_1,\dots,\tilde{b}_D),$$

2.2.2. Simplifications in the pairwise maximisation

The sum to maximise with respect to the parameters during the M-step is

$$\tilde{J} = \sum_{n=1}^{N} w_n \sum_{i<j} w_{ij} \log L_{Y2}((y_i^{(n)}, y_j^{(n)}); (\lambda_i^{(n)}, \lambda_j^{(n)}), \rho_{CY}^{(n)}). \tag{4}$$

As the parameters $\boldsymbol{\lambda}, \rho_{CY}$ are periodic, the sum can be simplified as

$$\tilde{J} = \sum_{t=1}^{T} \sum_{n,t_n=t} w_n \sum_{i<j} w_{ij} \log L_{Y2}((y_i^{(n)}, y_j^{(n)}); (\lambda_i^{(t)}, \lambda_j^{(t)}), \rho_{CY}^{(t)}). \tag{5}$$

No marginal memory

In case where there is no marginal memory; $\lambda_i^{(n)} = \lambda_i^{(t_n)}$. There are only 4 different possible values for the pair $(y_i^{(n)}, y_j^{(n)})$: $P_1 = (0,0)$, $P_2 = (0,1)$, $P_3 = (1,0)$ and $P_4 = (1,1)$. Then $\tilde{J}$ can be written as

$$\tilde{J} = \sum_{p=1}^{4} \sum_{t=1}^{T} \sum_{n, t_n = t, (y_i^{(n)}, y_j^{(n)}) = P_p} w_n \sum_{i<j} w_{ij} \log L_{Y2}(P_p; (\lambda_i(t), \lambda_j(t)), \rho_{CY}^{(t)}). \tag{6}$$

This reduces the computation of $ND(D-1)/2$ integrals to at most $4TD(D-1)/2$. Using the reformulation from Appendix 2.2.1, only one of the 4 situations has to be computed, as the others can be easily obtained from marginal probabilities. In the end, only $TD(D-1)/2$ computations are necessary.

With marginal memory

In the case with marginal memory, $\lambda_i^{(n)} = \lambda_{i, y_i^{(n-1)}}^{(t_n)}$. Similar simplifications allow to group observation pairs by 16 different possible values for $(y_i^{(n)}, y_j^{(n)}, y_i^{(n-1)}, y_j^{(n-1)})$.

This reduces the computation of $ND(D-1)/2$ integrals to at most $16TD(D-1)/2$. Using the reformulation from section 2.2.1, only $4TD(D-1)/2$ are needed.

2.3. Software

All implementation was performed in Julia (Bezanson et al. (2017)). For estimation purposes, the automatic differentiation package `Forwarddiff.jl` (Revels et al. (2016)) was used with the package `Optimization.jl` (Dixit and Rackauckas (2023)) and the LBFGS solver. The autodifferentiation-compatible implementation of the Matérn kernel function of the `BesselK.jl` package (Geoga et al. (2023)) was used. The integrals of the multivariate normal distribution and bivariate normal distribution are computed, respectively, using Julia package `MvNormalCDF.jl` for the Expectation step (with original code from Genz (1992)) and a custom implementation of Tsay and Ke (2023) for the Maximisation step. The Truncated Normal simulation method that is used is the Julia package `TruncatedMVN.jl` (Botev and Belzile (2015)), which uses the recent method of Botev (2017) rather than using acceptance-rejection techniques or Gibbs sampling, as implemented in the R package **tmvtnorm** (Wilhelm and G (2025)). Acceptance-rejection techniques are not suitable for high dimensions, and Gibbs sampling also quickly becomes computationally expensive.

3. Details on the French station data

Figure 3 (left) shows a map of the different locations, with the stations selected as illustrative examples highlighted in red. Their respective rain characteristics are shown in figure 3 (right). Table 1 gives for each location its ID, name, coordinates, elevation, and summary statistics related to their rain characteristics : rainy day frequency, maximum dry spell duration, maximum daily rainfall, mean daily rainfall. Theses statistics vary among locations, with locations close to the Mediterranean sea exhibiting longer dry spells but also much higher maximum rain. On the contrary, location in the North-West tend to have more frequent rain, with shorter dry spells, and more moderate maximum daily rainfall.

4. Details on the estimated parameters

4.1. Interpretability of the hidden states

4.1.1. State description

The results of the HMM parameter estimation allow the hidden states to be interpreted as weather regimes. The first state corresponds to an overall wet regime, the second to wet conditions in the North and dry conditions in the South, the third to wetter conditions in the South than in the North, and the fourth to dry conditions everywhere.

The states can be ordered according to the rain probability, from state 1 with the highest probability to state 4 with the lowest. This hierarchy is verified for almost all stations. The hierarchy of wettest to driest states is more pronounced for Lille (North, oceanic) than for the southern stations. For Nice in the

ID	Name	Lat	Lon	Elevation (m)	Rain frequency	Max dry spell duration (days)	Max daily rainfall	Mean daily rainfall
32	BOURGES	+47:03:33	+002:21:35	161	0.47	36	79.0	4.28
33	TOULOUSE	+43:37:16	+001:22:44	151	0.4	34	82.7	4.32
34	BORDEAUX	+44:49:50	-000:41:29	47	0.48	28	87.6	5.33
36	PERPIGNAN	+42:44:14	+002:52:22	42	0.25	50	222.0	5.87
736	ABBEVILLE	+50:08:10	+001:50:02	69	0.53	31	65.0	4.04
737	LILLE	+50:34:12	+003:05:51	47	0.51	33	62.8	3.93
738	CAEN	+49:10:48	-000:27:22	67	0.55	37	93.0	3.66
740	ALENCON	+48:26:44	+000:06:37	143	0.52	29	70.0	3.86
742	NANTES	+47:09:00	-001:36:32	26	0.5	35	96.7	4.44
745	DIJON	+47:16:04	+005:05:18	219	0.46	35	92.8	4.46
749	POITIERS	+46:35:38	+000:18:52	123	0.45	33	92.3	4.16
750	CLERMONT	+45:47:13	+003:08:58	331	0.4	37	76.0	3.97
786	MONTELIMAR	+44:34:52	+004:43:59	73	0.33	46	218.4	7.57
755	EMBRUN	+44:33:56	+006:30:08	871	0.31	51	67.2	6.55
756	TARBES	+43:11:17	+000:00:00	360	0.47	27	84.0	6.32
757	NICE	+43:38:56	+007:12:32	2	0.24	61	191.4	8.64
758	BASTIA	+42:32:26	+009:29:07	10	0.31	59	232.4	7.0
434	BREST	+48:26:39	-004:24:43	94	0.63	24	82.6	5.14
39	MARIGNANE	+43:26:16	+005:12:58	9	0.23	75	161.3	6.26
322	RENNES	+48:04:08	-001:44:02	36	0.5	29	70.4	3.77
323	STRASBOURG	+48:32:58	+007:38:25	150	0.47	32	66.3	3.68
2184	ROUEN	+49:22:59	+001:10:54	151	0.55	27	81.3	4.11
2190	TOURS	+47:26:40	+000:43:38	108	0.46	31	62.0	4.02
2192	BASEL	+47:36:52	+007:30:36	263	0.5	26	58.6	4.14
2195	LIMOGES	+45:51:40	+001:10:30	402	0.49	32	77.2	5.66
2199	GOURDON	+44:44:42	+001:23:48	260	0.46	33	102.6	5.05
2200	MILLAU	+44:07:07	+003:01:07	712	0.39	41	115.0	5.04
2203	MONT-DE-MARSAN	+43:54:35	-000:30:01	59	0.47	27	68.8	5.43
2205	ST-GIRONS	+43:00:19	+001:06:25	414	0.47	31	95.4	5.74
2207	MONTPELLIER	+43:34:37	+003:57:47	2	0.26	45	299.5	6.65
2209	AJACCIO	+41:55:05	+008:47:34	5	0.28	89	147.6	6.17
11244	CHASSIRON	+46:02:49	-001:24:42	11	0.47	32	59.6	4.01
11245	PLOUMANAC'H	+48:49:33	-003:28:23	55	0.59	22	51.4	3.93
793	LA HAGUE	+49:43:31	-001:56:23	6	0.52	39	52.5	3.96
11246	NANCY	+48:34:52	+005:57:35	336	0.51	36	80.0	4.36
11247	BELLE ILE	+47:17:40	-003:13:06	34	0.49	32	69.1	3.8
11249	ORLY	+48:43:01	+002:23:04	89	0.46	35	75.4	3.68

Table 1. Locations and characteristics of the 37 stations used in this paper.

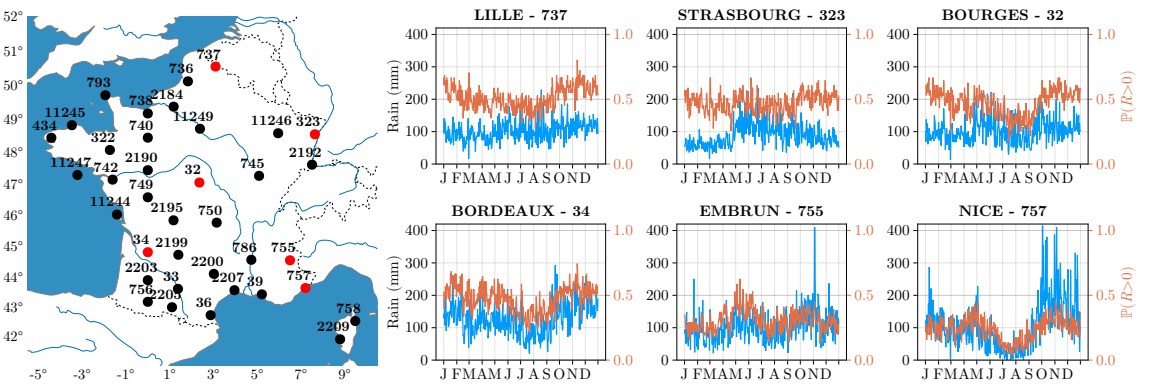

Figure 3: Left : map of the $D = 37$ weather stations used in this study. The six stations highlighted in red are used as illustrative examples in the validation section due to their contrasting climatic conditions. Right : mean daily rain amounts (blue) and daily rain probability (red) for each date of the year for the 6 locations chosen as examples.

South-East state 3 is as wet as state 1, with a high difference according to the previous day memory. In Embrun, a South mountainous location, the rainy state 1 and drier state 2 stay similar all year, but states 3 and 4 tend to be more rainy in the summer than in the winter. The opposite behaviour is observed in Lille, located in the north, where these states are relatively wetter in winter than in summer. For Strasbourg (North-East), Bordeaux (South-West), and Bourges (Central France), states 2 and 3 exhibit very similar

moderate rain probabilities. Despite this similarity across locations, their seasonal modulation differs, with a more pronounced summer decrease in rain probability at Bordeaux. These two states appear to overlap for these locations, while they show a clear different behaviour (state 2 rainier than state 3) for the other stations.

Transition probabilities provide useful information on the inferred states. States 1 and 4 are the most persistent, with a high probability of staying in the state. Moreover, it is almost impossible to go directly from state 1 (wettest) to state 4 (driest). States 2 and 3 act as transition states: from state 1 (or 4), there is a much higher transition rate to go to state 2 or 3 than to state 4 (or 1). From state 2, there is a relatively high probability to go to state 1 (and a lower probability, but non-zero, to go into 4). From state 3, it depends on the season: in spring, the most likely transition is into state 3; in autumn, states 2, 3, and 4 have similar probabilities.

The spatial scale also changes between states. State 4 (the driest) has a lower estimated scale, suggesting this state has shorter-range events. In contrast, states 1 (wettest) and 2 (wet north) have a higher spatial scale, indicating larger-range events.

4.1.2. Relationship with observed meteorological events

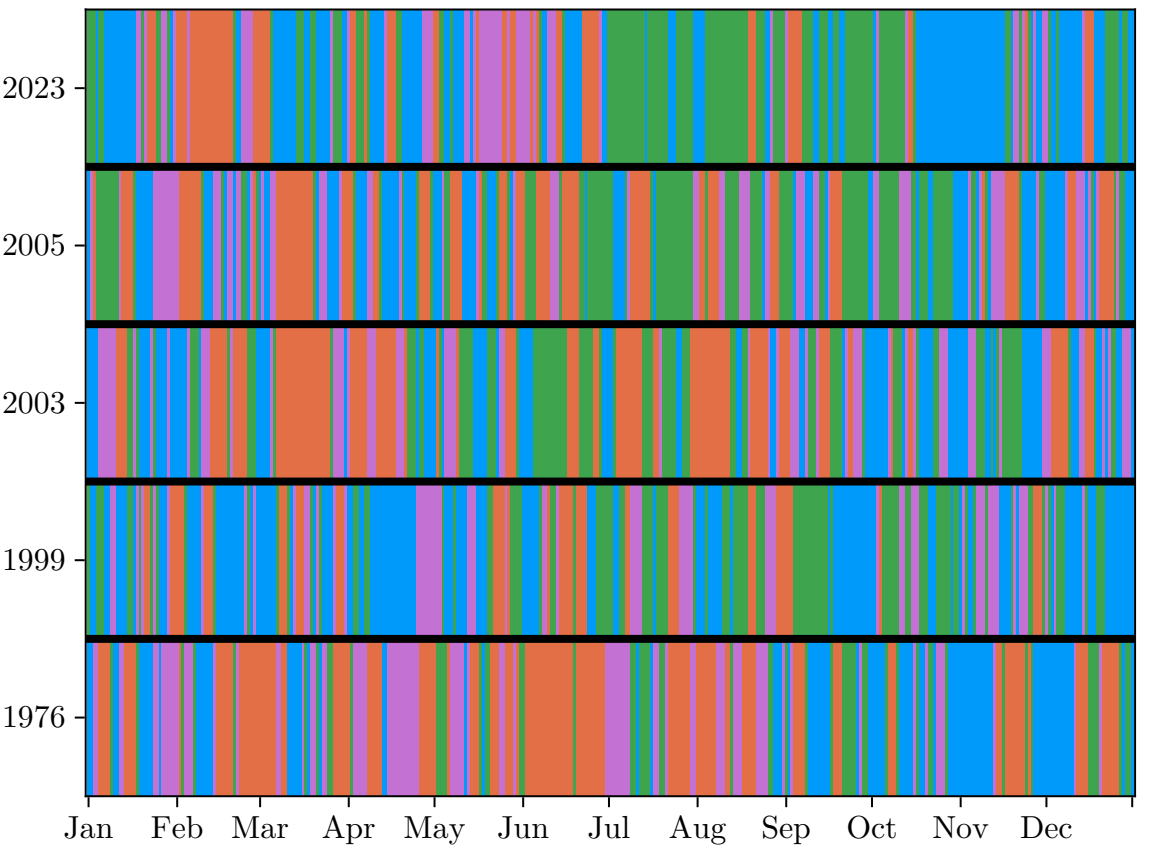

Figure 4: Most likely a posteriori sequence of states for several years. The wettest state 1 is in blue, state 2 (rainy north, dry south) in green, state 3 (rainy south) is in purple, and state 4 (driest) is in orange.

The four states can be illustrated by some remarkable meteorological events in the past. Figure 4 illustrates the most probable sequences of these states, colour-coded consistently with previous figures, for years 1976,

1999, 2003, 2005, and 2023. Major heatwaves are retrieved, as they correspond to dry events. Those occurring in August 2003 and during the summer of 1976 correspond to sustained episodes of the driest state (state 4), as anticipated. Both events were preceded by anomalously dry spring conditions, also captured in thegenerator by extended sequences of state 4. The year 1999 was "wet" according to the French climatology (see (SDES)). It exhibits a marked dominance of the wettest state (state 1) and comparatively few occurrences of state 4. In contrast, 2005 is identified as a dry year and shows infrequent and short-lived wet episodes, consistent with the limited representation of state 1. In 2023 (Météo-France (2024)), three periods show interesting behaviours. There was a precipitation deficit in the winter 2022-2023, with no rain between 21 January and 21 February, which is reflected by the persistence of state 4 over this period. In contrast, the end of the year experienced the longest rain episode since 1988, with a length only seen twice in the historical records, which is characterised by a very long episode of state 1. Perhaps more interestingly, the summer is characterised by a long period of the second state. This state expresses a North/South contrast, with wet north and dry south. Météo-France reported autumn-like conditions in the northern half of the country while record-breaking heat affected the south, consistent with the interpretation of this state.

4.2. Rain parameters

Figure 5 shows the estimated rain marginal parameters for the remaining 4 out of 6 stations not shown in the main text.

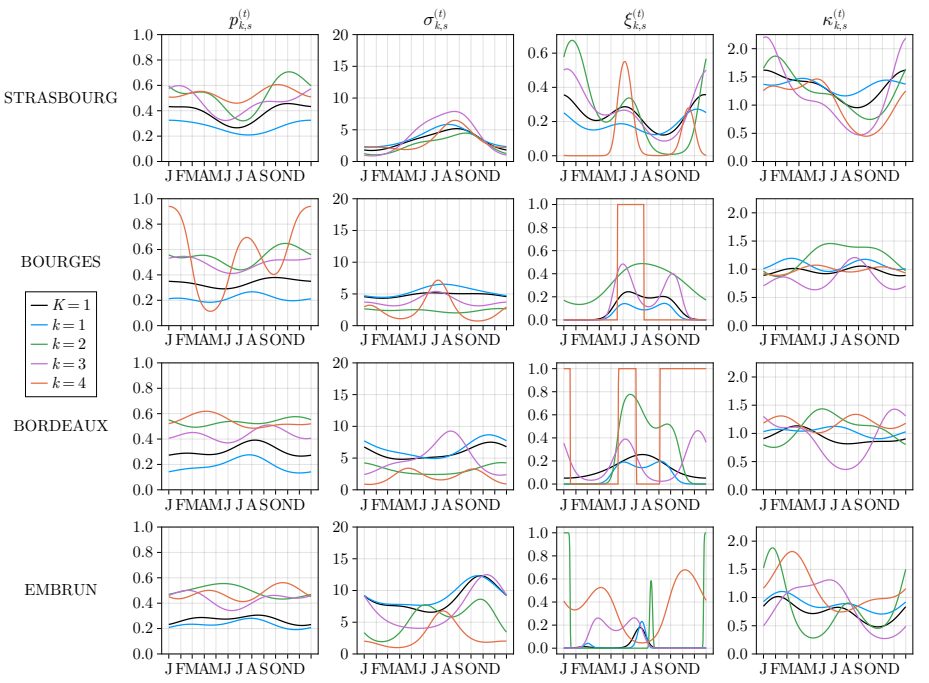

Figure 5: Yearly evolution of fitted rain intensity parameters for the 4 selected locations not shown in the main text. $p_{k,s}^{(t)}$ is the probability of low rain, $\sigma_{k,s}^{(t)}, \kappa_{k,s}^{(t)}, \xi_{k,s}^{(t)}$ are the EGPD parameters.

5. Additional indicators for the evaluation of the generator

5.1. Local dry spells

Figure 6 shows the distribution of local wet episode lengths for the 6 chosen locations. The red envelope represents the 25 to 75 % quantiles of the distributions in the simulation. For both types of events, the observations tend to be in the red envelope or to have slightly longer episodes. The gray envelope, covering all simulations, shows that simulations are able to generate longer events than those observed. These results are shared with dry event (see Supplementary material), which shows that thegenerator correctly reproduces local wet/dry events, even for lengths longer than 1 day.

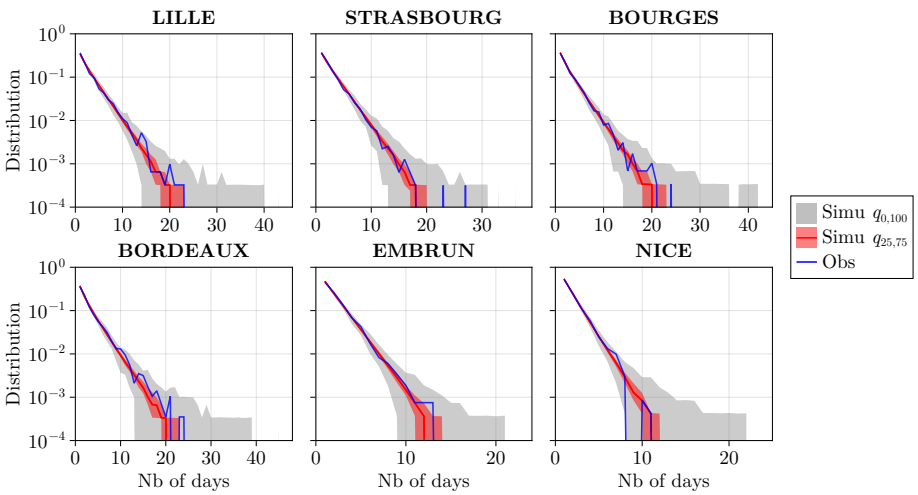

Figure 6: Distribution of temporal length of wet spells for the observations (blue line) and in 500 simulations (red and grey envelopes) for 6 locations.

5.2. 5-day cumulative rain

To complement the monthly analysis, the cumulative precipitation over consecutive 5-day periods is also examined, providing insight into short-term rainfall persistence and temporal dependence. For each location s and each day of the year t, the 5-day cumulative rainfall is computed from the daily intensities $R_s^{(n)}$ as

$$R_{s,n}^{(\text{5-day})} = \sum_{j=n-4}^{n} R_s^{(j)}, \tag{7}$$

using a moving 5-day window over the full annual cycle. Compared to daily precipitation, this aggregated indicator smooths out day-to-day noise and reflects the occurrence of multi-day rainfall episodes, which are crucial for hydrological applications.

Figure 7 shows the cumulative 5-day characteristics for each date of the year as defined in the main text. The blue ribbons represent the 90% quantile of that day of the year, green ribbons the median of that day, and orange ribbons the 10% quantile for that day, with dark blue lines representing the observed values. The low quantile is almost always 0 and is not visible on the plot. For locations like Nice or Embrun, the 5-day cumulative rain is mostly close to 0, with the high-quantile days being responsible for most of the monthly precipitation, denoting a low temporal dependence. For the other locations, the median 5-day precipitation is higher. In all cases, the simulation envelopes contain the observations.

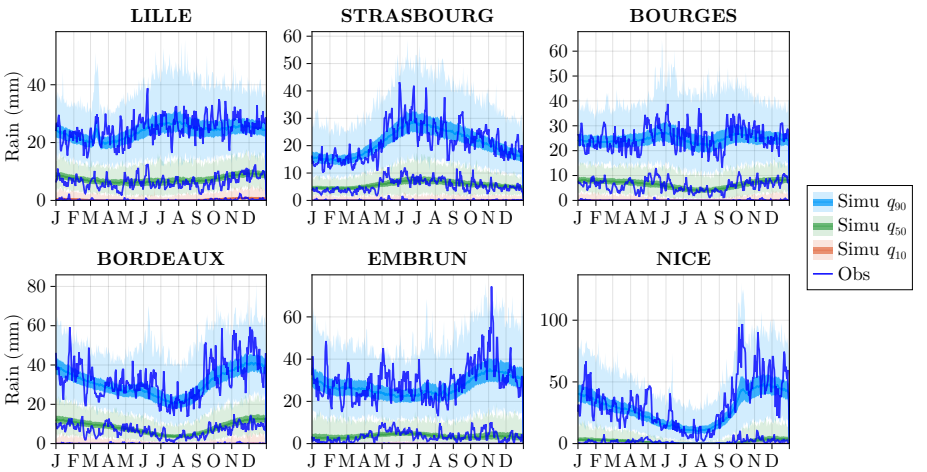

Figure 7: 5-day cumulative rain for selected locations. 0.1 quantile in orange, 0.5 quantile in green, 0.9 quantile in blue. The lowest quantile is mostly 0, which makes it not visible. Each simulated envelope includes a central ribbon corresponding to the interquartile range (0.25–0.75 quantiles) and an outer ribbon showing the full range of the simulations (minimum–maximum).

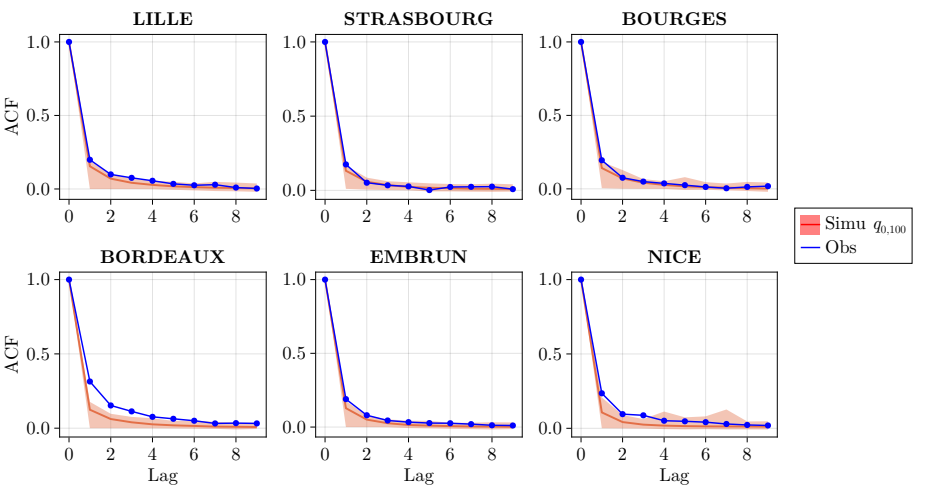

Figure 8: Auto-correlation function of the precipitation intensity for selected locations, for time lags between 0 and 10 days. The values from the observations are in blue, and the simulated envelope is in red.

5.3. Temporal dependence in the simulated rain amounts

Figure 8 shows the auto-correlation function of the rain intensity for the same 6 locations for observations and simulations. The observations in blue lie mostly within the gray simulation envelope, although thegenerator tends to underestimate the temporal correlation, especially for Bordeaux or Nice.

5.4. Probability of joint rain occurrence

Figure 9 shows joint probabilities of rain occurrence between pairs of stations, for the four seasons. Observed values on the x-axis are compared with simulated values on the y-axis and are found to be aligned on the 1:1 line, both for same-day probabilities and for lag-1 probabilities.

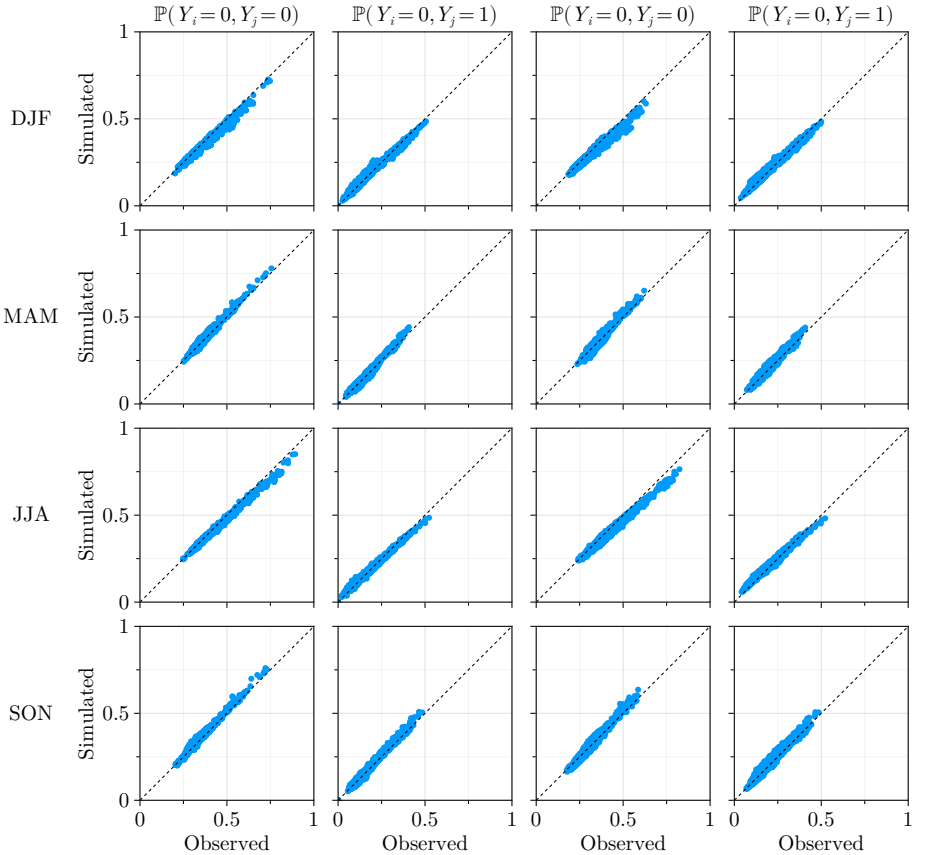

Figure 9: Joint rain occurrence probabilities on the same day (columns 1 and 2) and on consecutive days (columns 3 and 4), for the different seasons DJF: Winter, MAM: Spring, JJA: Summer, SON: Autumn, and all pairs of sites.

5.5. Spatial dry spell durations

Figure 10 shows the distribution of dry episode lengths for the observations and the 4 occurrence models, for a dry day threshold set to 10%. The results are similar to the less extreme threshold of 20%, however, a small underestimation of the lengths of dry episodes can be seen in the summer.

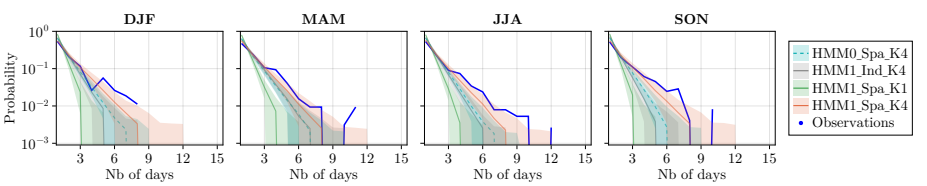

Figure 10: Distribution of length in days of dry spatial episodes, defined as $\mathrm{ROR}(n) < 0.1$ for the observations and 4 generators. The values in the observations are the blue curves, and each coloured ribbon corresponds to the min-max envelope from simulation of the generators HMM0_Spa_K4, HMM1_Ind_K4, HMM1_Spa_K1, HMM1_Spa_K4.

5.6. Relaxed ROR definition

Using an indicator that treats very small rainfall amounts as dry days is more relevant for drought risk assessment than defining a dry day strictly as one with zero precipitation. Therefore, a threshold of $R > 1\,\mathrm{mm}$ can be chosen when defining a wet day, and the $\mathrm{ROR}_{1.0}(n)$ defined as

$$\mathrm{ROR}_{1.0}(n) = \frac{1}{D}\sum_{s=1}^{D} 1_{\{R_s^{(n)} > 1.0\,\mathrm{mm}\}} \tag{8}$$

This indicator of dry days can be correctly represented only if the occurrence process Y and the rain intensity process R are properly linked, as both low rainfall amounts and zero rainfall must be jointly captured. Figure 11 shows that the generator successfully reproduces this indicator. The 1.0 mm tolerance increases 3 times the proportion of "dry" days.

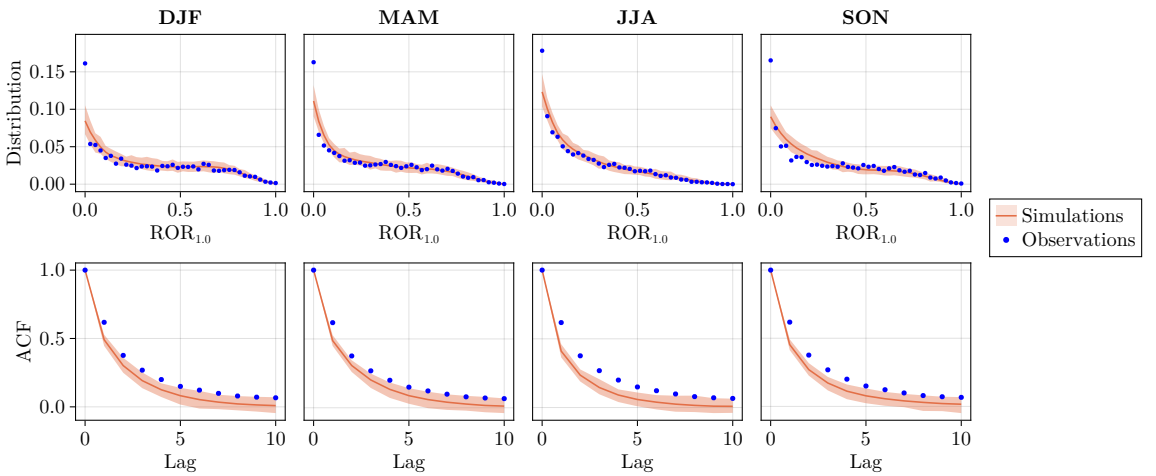

Figure 11: Distribution (first row) and auto-correlation function (second row) of $\mathrm{ROR}_{1.0}$ defined as the rain occurrence ratio with a tolerance of 1mm. The simulations from thegenerator are represented by its min-max envelope in orange, and the observed values are the blue dots.

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